

# Condensate Partition Coefficient Pipeline

## Instruction Manual

Automated 3D segmentation of nuclear biomolecular condensates and measurement of the partition coefficient from confocal Z-stacks

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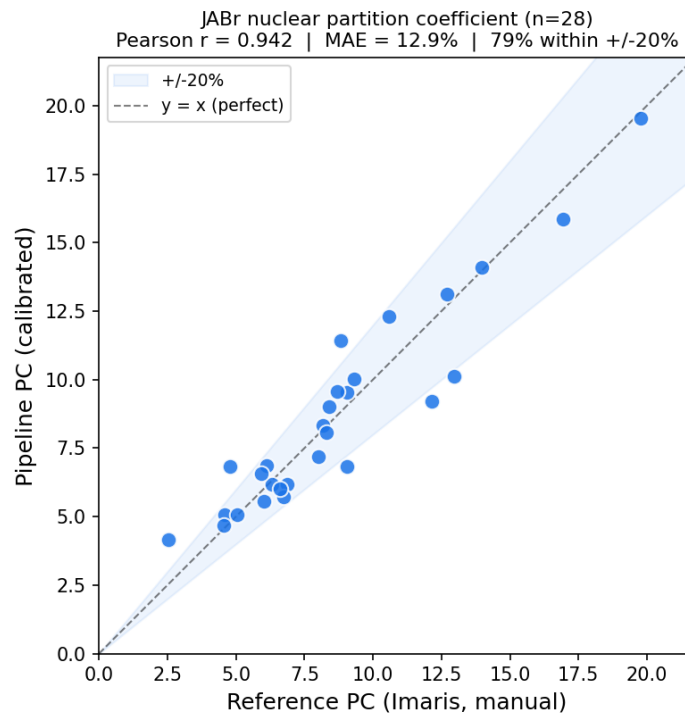
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## Overview

This pipeline replaces the manual Imaris workflow for measuring the **partition coefficient (PC)** of nuclear biomolecular condensates. Given a two-channel confocal Z-stack, it segments nuclei and condensates in 3D, computes the PC, and reports a calibrated value on the manual reference scale.

| Item         | Description                                                                         |
|--------------|-------------------------------------------------------------------------------------|
| Inputs       | Two-channel Z-stack: channel 0 = nuclei, channel 1 = condensate.                    |
| Outputs      | Nuclear & cytoplasmic PC, 3D masks, per-object volumes, summary figure.             |
| Validated on | JABr: Pearson $r = 0.942$ , MAE 12.9%, 79% of cells within $\pm 20\%$ ( $n = 28$ ). |
| Interfaces   | Graphical (run_gui.py) or command line (pipeline.py).                               |



JABr validation: calibrated pipeline PC vs manual Imaris reference.

## Installation (one time)

1. Install Python 3.10 or 3.11.
2. Install PyTorch for your machine from [pytorch.org](https://pytorch.org) (CPU or your CUDA version), e.g. CUDA 12.1:

```
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

3. Install the remaining dependencies:

```
pip install -r requirements.txt
```

On first run, Cellpose downloads its model weights automatically (one-time ~100 MB). A GPU is recommended (~10–15 s per cell); CPU works but is slower.

## Preparing your image files

The pipeline accepts either one multi-channel TIF (both channels in one file) or two separate single-channel TIFs. **Channel order matters:** channel 0 must be nuclei, channel 1 the condensate. Most axis orderings (ZCYX, CZYX, ...) are auto-detected. If results look wrong, swapped channels are the most common cause.

## Running with the graphical interface

Launch with `python run_gui.py`. A window with **Single File** and **Batch** tabs opens.

### Step by step (single cell)

1. On the **Single File** tab, choose the input mode and Browse to your data.
2. Choose an Output folder (optional).
3. In **Settings**, set **Construct = JABr**; leave all other settings at their defaults.
4. Click  **Run Pipeline**. Progress streams in the Output Log; the status reads **Done** when finished.
5. Open your output folder for the masks, tables, and results.png.

Live GUI screenshots should be captured on the machine where it runs (Windows: Win+Shift+S) and saved into docs/figures/. The actual numerical output of a real run is shown under “Reading the outputs.”

## Understanding the settings

For routine JABr analysis you only need to set **Construct**; the rest have sensible defaults.

| Setting              | Default | What it does                                                                                       |
|----------------------|---------|----------------------------------------------------------------------------------------------------|
| Construct            | (none)  | Pick JABr for the validated workflow. Auto-selects the blob_log detector and the JABr calibration. |
| Condensate detector  | auto    | auto routes JABr→blob_log, else Cellpose. Normally left on auto.                                   |
| Top-X% brightest     | 75      | Defines condensed-phase density; trims the dim mask halo to match the manual reference.            |
| Nuclei cell-prob.    | -2.0    | How aggressively nuclei are detected. -2.0 suits the lab's stain.                                  |
| blob_log threshold   | 0.03    | Condensate spot sensitivity (used when detector = blob_log). Leave at 0.03 for JABr.               |
| Cellpose cond. model | (blank) | Advanced; a fine-tuned condensate model. Leave blank for JABr.                                     |
| Disable GPU          | off     | Tick on a laptop with no NVIDIA GPU.                                                               |

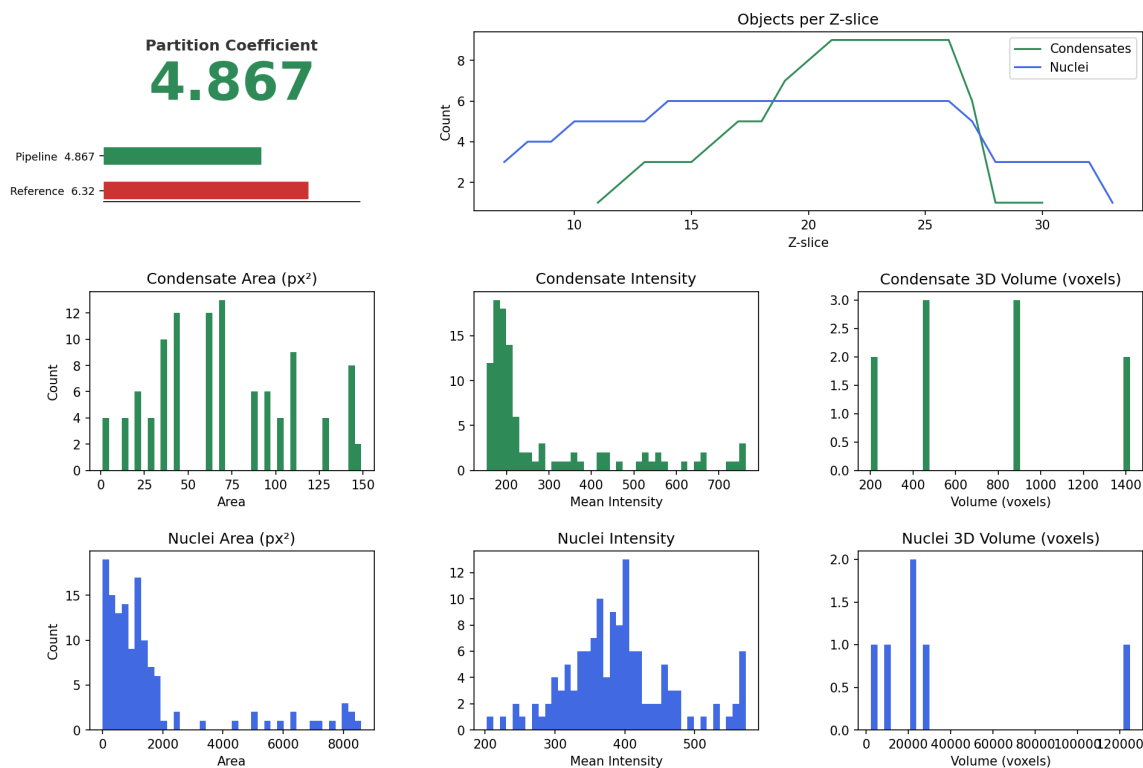
**The one rule for routine use:** set Construct = JABr and run. Everything else is for experimentation or other constructs.

## Reading the outputs

Every run writes summary.csv (headline numbers incl. the calibrated PC), results.png (summary figure), condensate\_masks.tif and nuclei\_masks.tif (3D labeled masks), and per-object/per-slice CSVs.

For the bundled sample JABr\_Sample2\_5\_3.tif, a real run produces **nuclear PC raw 4.867** → **calibrated 4.813** (manual reference 4.558, +5.6%).

**Report the calibrated nuclear PC.** The raw value is systematically ~3× the manual scale by construction; calibration standardizes it. **Always glance at condensate\_masks.tif** over the raw channel in Fiji/ImageJ — a 10-second check catches the rare misfire.



results.png from the bundled-sample run.

## Batch mode and command line

On the **Batch** tab, select a folder of .tif files; every file is processed. Optionally supply the manual Imaris nuclear-PC CSV as a Reference CSV — the pipeline then writes comparison.csv and scatter.png with the correlation, RMSE, and MAE.

Command line:

```
python pipeline.py --roi sample_data/JABr_Sample2_5_3.tif --construct JABr --output my_output
```

Separate channels: --nuc nuclei.tif --cond condensate.tif. Physical volumes in  $\mu\text{m}^3$ : add --voxel-xy 0.065 --voxel-z 0.3. Force CPU: --no-gpu.

## Troubleshooting

| Symptom                      | Likely cause / fix                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------|
| PC wildly off / nuclei empty | Channels swapped: confirm ch0 = nuclei, ch1 = condensate.                                                    |
| "0 condensates detected"     | Not JABr, or very dim signal. Lower blob_log threshold (e.g. 0.02); confirm the raw channel has s            |
| Very slow (minutes/cell)     | Running on CPU. Install a CUDA PyTorch build; ensure Disable GPU is unchecked.                               |
| CUDA out of memory           | Tick Disable GPU, or close other GPU programs.                                                               |
| Cellpose download fails      | First run needs internet for model weights.                                                                  |
| One cell's calibrated PC off | Calibration is population-level; cells vary ( $\pm 20\%$ for $\sim 80\%$ of cells). Verify against the mask. |

## Method summary and scope

The PC is the ratio of condensed-phase to dilute-phase density (Fabrini et al.), both camera-background-subtracted. Condensed density uses the brightest 75% of voxels inside (condensate  $\cap$  nucleus); dilute density is the mean of the 50 lowest-intensity  $10 \times 10 \times 10$  patches in the nuclear dilute region. Nuclei are segmented with Cellpose 3 (cyto3, 3D), cleaned, and void-filled; condensates are detected with Laplacian-of-Gaussian blob\_log and kept when  $\geq 50\%$  of the rendered sphere lies inside a nucleus. A per-construct linear (JABr) or isotonic calibration standardizes the automated PC onto the manual scale.

**Scope.** Validated and production-ready for JABr. The detector is construct-specific; a leave-one-construct-out test confirmed a single detector does not generalize zero-shot (it under-detects on unseen constructs). Extending to other constructs needs per-construct calibration or few-shot retraining. Full rationale and the cross-construct evidence are in docs/METHODS.md.